METHOD FOR LEGACY PROTOCOL TUNNELING OVER A HIGH-CARDINALITY INTERFACE

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Title: METHOD FOR LEGACY PROTOCOL TUNNELING OVER A HIGH-CARDINALITY INTERFACE

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Abstract

A method for transmitting data encoded with a legacy, human-readable character set (e.g., UTF-8, UTF-16) over a high-cardinality, massively parallel communication interface. The method comprises: creating a specific encoding map that provides a one-to-one mapping between each character in a pre-defined legacy character set and a unique state of a single physical channel on the parallel interface. This allows for the transmission of one full character per channel per frame. This "retrofit" method enables existing text-based protocols and data formats to be tunneled over the new physical layer at extremely high speeds, treating entire characters as atomic units of transmission rather than sequences of bits.

Background of the Invention

Legacy data formats and protocols, particularly those designed for human-readable text such as UTF-8, are ubiquitous. While new high-cardinality interfaces are optimized for

dense, machine-readable encodings, there is a significant practical need to support these legacy formats for backward compatibility and interoperability. Transmitting UTF-8 data by breaking each character down into its constituent bits and sending them serially would be highly inefficient and would fail to leverage the parallel nature and high state-space of the new interfaces. A method is needed to transport these legacy encodings more efficiently.

Summary of the Invention

The present invention is a method for "tunneling" or retrofitting legacy character sets, such as UTF-8, for transmission over a high-cardinality, parallel communication interface. The method provides a bridge between old data formats and the new physical layer, enabling dramatic performance improvements without requiring changes to the legacy data itself.

The core of the method is the creation and use of a specialized encoding map. This "legacy retrofit" map provides a direct, one-to-one correspondence between every character defined in a given legacy character set (e.g., the 1,112,064 possible code points in Unicode) and a unique state of a physical channel (e.g., a unique pixel color). A transmitter, upon receiving a stream of text, does not break the characters into bits. Instead, for each character, it looks up the corresponding channel state in the legacy retrofit map and sets a single physical channel to that state. The receiver uses the same map to perform the reverse process, instantly decoding each channel state back into the original character. This allows an entire text document to be transmitted in a fraction of the time required by traditional methods.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a diagram contrasting traditional bit-serial transmission of a legacy character with the method of the present invention.

Detailed Description of the Invention

Referring to **FIG. 1**, a comparison is shown for transmitting the UTF-8 character 'G' (binary 01000111). In the traditional method (100), the character's data (101) is broken into a sequence of eight individual bits (102), which are transmitted serially over a low-cardinality channel (103). The method of the present invention (110) operates differently. The character 'G' (101) is processed by an encoder using a legacy retrofit map (111). The map provides a direct mapping from the character 'G' to a single, unique state of a high-cardinality physical channel (112), for example, a specific shade of blue. This single state is transmitted in one step over the parallel interface (113).

Claims

What is claimed is:

1. A method for transmitting data over a parallel communication interface having a plurality of physical channels, each capable of representing N unique states, the method comprising the steps of:

    - providing a pre-defined legacy character set containing a plurality of characters;
    - creating an encoding map that maps each character of the legacy character set to a unique state of the N states;
    - receiving an input data stream comprising a sequence of characters from the legacy character set; and
    - for each character in the sequence, transmitting it by setting a single physical channel to the corresponding unique state defined in the encoding map.

2. The method of claim 1, wherein the legacy character set is a Unicode standard, such as UTF-8 or UTF-16.

3. The method of claim 1, wherein the physical channels are pixels and the N unique states are pixel colors.

4. The method of claim 1, wherein a plurality of characters are transmitted simultaneously in a single frame by using a plurality of physical channels.

5. The method of claim 1, wherein the parallel communication interface is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.

Drawings

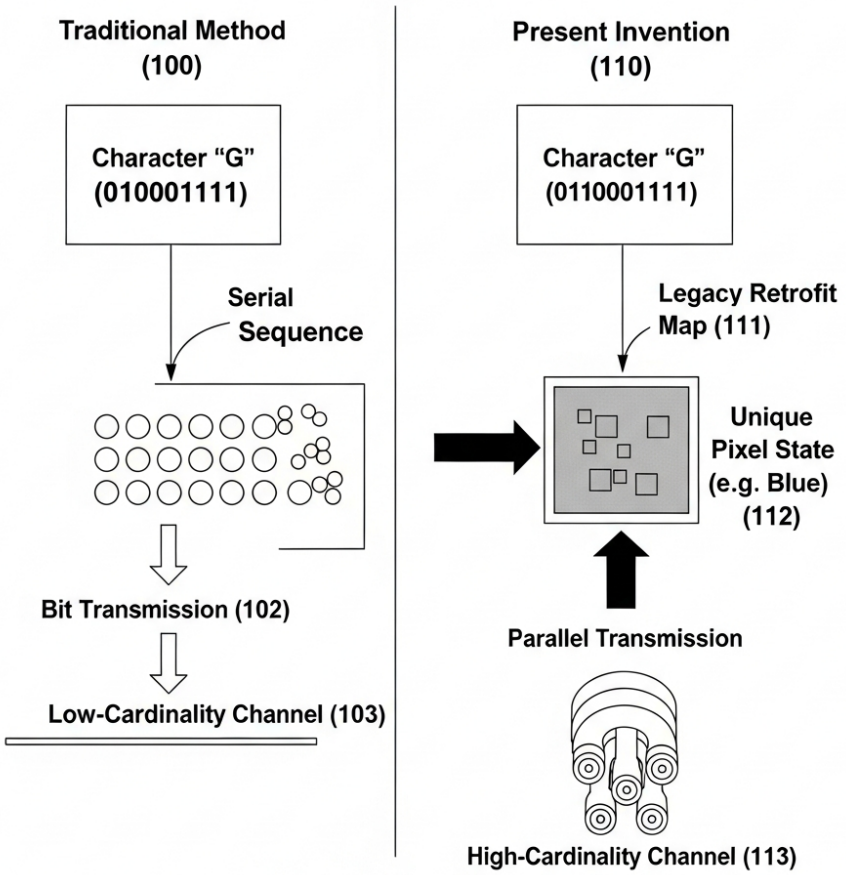


Figure 1: Diagram contrasting traditional bit-serial transmission of a legacy character with the method of the present invention.